M321: Polynomial Thin Trees

Rational Certificates, Bit Complexity, and Conditional Rounding Improvements

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Abstract

We examine the complete mathematical source of *A polynomial-time construction of strong thin trees* at the pinned OpenAI public snapshot. The main issue is not the fractional scale, but obtaining one integral tree with simultaneous cut control and a polynomial binary-time construction. We rederive the continuation, normalization, padding, signing and iteration interfaces, with particular attention to exact rational separation, lower spectral margins, fixed denominator grids and zero-dimensional cases. We prove a whitening-free rational certificate for a single subgraph that simultaneously controls every cut and retains a tree packing. Additional local results quantify coefficient approximation and improve the compressed input transfer factor from 2 to at most $8/7$. The same integer rounding estimate yields a conditional $96B/7$ thin-and-cheap bound in place of $24B$. These are local implications, not a new proof or computational validation of the universal thin-tree theorem. The geometric hierarchy transfer and the constructive signing theorem remain explicit global dependencies. The private thin-tree report supplies exact formulations and certificate discipline, not those missing construction theorems. A source ledger and an independently executable exact checker delimit the evidence.

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1 Scope, quantifiers and evidence status

1.1 The pinned claim

Let G be a finite undirected loopless multigraph, with labelled copies when edges are explicit. For $S \subseteq V$, write $\delta_G(S)$ for the multiset crossing S. The source claims a universal constant C and a deterministic algorithm such that

$$\begin{aligned} &\forall n \geq 1\ \forall k \in \mathbb{Z}_{\geq 1}\ \forall G, \quad \lambda(G) \geq k \implies \\ &\exists T \text{ spanning tree of } G\ \forall \varnothing \neq S \subsetneq V, \quad |\delta_T(S)| \leq C|\delta_G(S)|/k. \end{aligned} \tag{1.1}$$

The algorithm is claimed polynomial in the total binary input length, both for explicit edges and for endpoint pairs with binary multiplicities. The word "one" in the existential tree quantifier is essential. Neither different trees for different cuts nor a per-cut expected bound proves (1.1). The source is the September 23, 2026 manuscript at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a` [1]. The snapshot metadata date is October 6; this review date is October 7.

We use three evidence classes. *Source-claimed* means a statement in the manuscript, including its global construction. *Locally proved* means an implication with a complete proof here. *Tested* means only the finite cases enumerated by the accompanying checker. These classes are not interchangeable. No full implementation of the ellipsoid/signing pipeline, benchmark, Lean replay or universal validation was performed. The catalogue's Level 5 records direct applicability to the same target; it is not a validity, novelty or performance grade.

1.2 Boundary cases and representation

For $n = 1$ return the empty tree before any inverse, partition oracle or packing call. Cut inequalities are then vacuous. The source excludes $n = 0$; an implementation should reject that input rather than infer a spanning-tree convention from an empty cut family. For $n \geq 2$, $k \geq 1$ and the connectivity promise imply connectedness. For $k = 1$ use one tree; the average of $\lfloor k/2 \rfloor$ trees is not defined. This is a domain qualification needed for the fractional theorem in the private report, whose formula involving $1/\lfloor k/2 \rfloor$ must be read with $k \geq 2$.

Loops may be discarded in the fractional extension because their costs are nonnegative and they cross no cut. Zero multiplicities are omitted; zero fractional coordinates cannot supply a returned edge. A tree cannot contain two parallel copies of the same unordered pair: they form a cycle of length two in the graphic matroid. This observation, not an injective lifting of the entire auxiliary graph, justifies the final lift. With explicit records and a connected positive support, at least $n - 1$ distinct nonloop pairs are present, so n is bounded by the input length even if its declaration is binary.

1.3 What was read and what remains conditional

All thirteen files under the assigned `build/source` directory were read, including all nine section files (among them the separately included main theorem), figure source, macros and bibliography. The direct private report TR-item-224-astra was read in full. Relevant map entries were followed to actual frame and coordinate-transfer theorem passages; maps were not used as substitutes for their proofs. The exact files and hash evidence are specified in Section 11.

The geometric section imports the companion's bag extraction, pruning and potential accounting. Reading its transfer argument does not amount to re-proving every lemma in that different manuscript. Consequently the global hierarchy theorem, and hence the existence of all continuation responses,

is retained as a named dependency. The signing appendix was read in full and its quantitative interfaces are examined below; its entire analytic algorithm was not independently executed. No conclusion here says that a finite checker settles either global dependency.

2 The private bridge and the public baseline

2.1 What the private report establishes

TR-item-224-astra [3] separates qualitative Goddyn thinness from the rate C/k. The latter implies the former; an arbitrary connectivity threshold function does not give that rate. For $k \geq 2$, Nash–Williams–Tutte gives $r = \lfloor k/2 \rfloor$ disjoint trees. Indeed, for a partition $\mathcal{P}$ into $d \geq 2$ parts,

$$|E_G(\mathcal{P})| = \tfrac{1}{2} \sum_{P \in \mathcal{P}} |\delta_G(P)| \geq kd/2 \geq r(d-1).$$

The one-part partition is treated separately and has zero crossing edges. For $x = r^{-1} \sum_j \mathbf{1}_{T_j}$, edge-disjointness gives $x_e \leq 1/r$ and therefore $x(\delta S) \leq |\delta_G(S)|/r$ for every cut. This is a fractional conclusion, not a choice of one suitable T_j.

With $b_{uv} = e_u - e_v$ and $L_F = \sum_{e \in F} b_e b_e^T$, the report's exact and relaxed quantities are

$$\alpha_{\text{cut}} = \max_S \frac{\mathbf{1}_S^T L_T \mathbf{1}_S}{\mathbf{1}_S^T L_G \mathbf{1}_S}, \qquad \alpha_{\text{spec}} = \sup_{0 \neq z \in \mathbf{1}^\perp} \frac{z^T L_T z}{z^T L_G z}.$$

Centering cut indicators proves $\alpha_{\text{cut}} \leq \alpha_{\text{spec}}$. The denominator is positive only after connectedness and removal of the constant direction. The coarea identity

$$\sum_{uv} w_{uv} |z_u - z_v| = \int_{\mathbb{R}} w(\delta\{v : z_v > t\})\, dt$$

gives an exact ℓ^1 reformulation of cut domination. Replacing the absolute values by squares changes the problem. The private report also proves $\alpha_{\text{spec}} \leq (n-1)\alpha_{\text{cut}}$ by sorting coordinates and applying Cauchy–Schwarz to the nonnegative consecutive gaps; this is not a dimension-free transfer.

Positive domination of all cut-incidence vectors by a protected family, with nonnegative coefficients and a bounded cut-size budget, transfers a shared-tree bound by pairing with $\mathbf{1}_T$. Signed uncrossing alone is not such a certificate. The report's tree-path parity representation is complete but exponential: $2^{n-1} - 1$ nonempty masks parameterize the cuts up to complementation. Restricted masks give lower bounds on worst-case thinness, not all-cut upper bounds. These distinctions are preserved.

The report correctly warns that the uniform mixture of a packing need not be negatively correlated. Its K_5 example has $\Pr(01, 12 \in T) = 1/2 > 1/4$. A weighted-tree law with the same marginals is a different construction. Nothing in our certificates assumes a negative-dependence property for a packing mixture.

2.2 Norms and coordinates from the corpus

The actual TR248 finite-model and perturbation results [4] require a specified physical space and lower frame bound. If $AI \preceq T^*T$ and $\|E\| \leq \eta < \sqrt{A}$, then

$$\|(T+E)x\| \geq (\sqrt{A} - \eta)\, \|x\|, \qquad (T+E)^*(T+E) \succeq (\sqrt{A} - \eta)^2 I.$$

The proof is the triangle inequality. Exact rank information must not be silently assigned to rounded columns merely because their errors are small. TR250's compatible-transfer theorem [5] uses one invertible map and transports the physical Gram matrix by congruence. It preserves kernels and energies; it creates neither connectivity nor improved thinness. These are concrete reasons to retain exact positive weights and a common invertible whitening map in M321.

2.3 Public literature refreshed on the review date

Anari–Oveis Gharan's public full version provides the general $\mathrm{poly}(\log\log n)/k$ existence scale and the hierarchy framework [6]. Klein–Olver treats a prescribed laminar family [7]; Klein–Olver–Yeoh treats cuts strictly below $(1+1/40)k$, with the stated bound 88 crossings [8]. These restricted or weaker bounds must not be described as the all-cut inverse-connectivity construction asserted by M321.

The public Ezeunala–Jiang version 1, dated September 15, states a deterministic rational rank-one discrepancy theorem with constant 13 [9]. M321 uses a conservative constant 40 and supplies a longer precision argument. The public proof's compactness sentence, by itself, does not justify a polynomial derivative bound uniformly in input size; the quantitative Jacobian argument in M321 is therefore substantive, not optional. The covariance input is explicitly supplied by the Bansal–Garg/Bansal results [10].

Terao's matroid-partition result supplies a polynomial independence-oracle implementation [11]. Acyclicity is an exact graphic independence oracle, and the number of matroid copies is charged to the explicit edge count. MSS supplies an existential partition theorem [13], not by itself the deterministic algorithm. No benchmark or priority claim is inferred from this literature comparison.

3 Geometry, cone direction and shared responses

All matrices in the geometric argument act on $\mathsf{E} = \mathbf{1}^{\perp}$ with its inherited Euclidean norm. Set

$$\mathcal{N} = \{Z \succeq 0 : b_{uw}^T Z b_{uw} \le b_{uv}^T Z b_{uv} + b_{vw}^T Z b_{vw}\},$$
$$\mathcal{K}_H = \{X \succeq L_H : \langle X, Z\rangle \le 3\langle L_H, Z\rangle \ (Z \in \mathcal{N})\}.$$

The centered identity and every centered cut Gram matrix belong to $\mathcal{N}$. Enlarging a test cone makes its dual inequality stronger: a shortcut valid only on cut indicators is not automatically valid on $\mathcal{N}$. Conversely $X \in \mathcal{K}_H$ gives $X \succeq L_H$ in Loewner order but only a *cone-tested* upper bound. It does not assert $X \preceq 3L_H$ on all vectors.

3.1 Two elementary geometric reductions

For $A, Z \succeq 0$,

$$\inf_{D \succ 0}\{\mathrm{tr}(AD^{-1}) + \mathrm{tr}(ZD)\} = 2\,\mathrm{tr}((Z^{1/2}AZ^{1/2})^{1/2}). \tag{3.1}$$

For positive definite matrices put $H = Z^{1/2}DZ^{1/2}$ and $C = Z^{1/2}AZ^{1/2}$. The difference from the right side is $\left\|H^{-1/2}C^{1/2} - H^{1/2}\right\|_F^2$, minimized at $H = C^{1/2}$. Adding ηI to A, Z and taking $\eta \downarrow 0$ proves the singular case by the two opposite infimum bounds. The source uses this identity after strict conic feasibility to convert the shortcut objective into a ratio of squared boundary sums to total squared edge length. It does not assume attainment of the original shortcut infimum.

The doubled profile construction is also exact. For a finite metric d choose $R > 2\,\mathrm{diam}\,d$ and give two copies of its points distances $d(x,y)$ within a copy and $R - d(x,y)$ between copies. Triangles

with two cross-copy sides reduce either to a triangle of d or to $d(x,y)+d(y,z)+d(z,x)<2R$. Let c_a be the distance profile of doubled point a and $Q=\operatorname{conv}\{c_a\}\subset\ell_\infty$. For $q=\sum_b\theta_b c_b$, the triangle inequality gives $\|q-c_a\|_\infty\le\sum_b\theta_b\widehat{d}(a,b)$ and coordinate a gives equality. Thus every radius is affine, opposite radii sum to R, and motion toward either endpoint changes radius by exactly the distance travelled. For a pseudometric only zero-distance geometric images are identified: graph vertices and parallel-edge resources remain distinct. This prevents an unjustified graph contraction.

3.2 The hierarchy dependency

The source's hierarchy estimate bounds marked average resistance by $h^{-31/40}$ when parent boundaries satisfy $|O(A)|\ge|P(A)|/h$. The transfer proof retains the companion's scalar accounting but changes the continuous geometry. Its essential distinctions are ordinary balls versus punctured shells, geometric disjointness versus disjoint hierarchy conflict domains, and inner witnesses versus membership in a shell. Removing a center need not remove a witness, but it changes membership tests. The assigned-path argument charges disjoint open portions of an edge, with each parallel copy a separate resource.

The displayed parameter arithmetic is consistent:

$$\alpha=h^{-4/5},\quad \epsilon=1/10,\quad \sigma=h^{-4/5}/C_2,\quad C_2=O((1+\log h)^4),$$

so $h\sigma^{1+2\epsilon}=h^{1/25}/C_2^{6/5}\to\infty$. The total dual ratio is bounded by $2h^{-4/5}+O(h^{-23/25}(1+\log h)^7)=o(h^{-31/40})$. The two power margins are $1/40$ and $29/200$. These scalar calculations do not independently prove the imported bag selection and charging assertions.

For a nested chain, the dyadic representative construction has total crossing counts $\sum_{j\ge i}M_j<4M_i$. Its one common response $Y=L_H+D$ lies in $\mathcal{K}_H$. Markov's inequality and boundary multiplicity at most two give a bad-edge fraction bounded by

$$16\big(h^{-1}+6sh^{-31/40}\big)=O(r^{-1}+r^{-21/40})\le r^{-1/3},\qquad s\le r^{1/4}.$$

The final exponent margin is $23/120$. The same Y for the whole chain is indispensable to the convex continuation; separate responses would not contradict a separated weighted obstruction.

4 Extraction with charged rational complexity

4.1 Ramp, uncrossing and continuation

Let H be explicit with m labelled edges and r disjoint trees. Put $\kappa=(m+1)^{-1}$, $t=\kappa/32$, and choose dyadic $r^{-1/8}\le g\le 2r^{-1/8}$ by integer eighth-power comparisons. At (s,X) define

$$R_e=b_e^TX^{-1}b_e,\quad p_e=\min(2,(sR_e)^{-1}),\quad u_e=\max(0,p_e-1).$$

The invariant is $(u+\kappa)(E_H(\mathcal{P}))\ge Q(|\mathcal{P}|-1)$. It holds at $s=1/2, X=L_H, Q=r$, because $R_e(L_H)\le 1$. For an active edge impose $Y\succeq 6sy_eb_eb_e^T$, $0\le y_e\le 1$, and update $s^+=(1+t)s$, $X^+=(1-t)X+tY$. The rank-one inverse-order equivalence gives

$$\frac{1}{s^+R_e(X^+)}\ge\frac{(1-t)p_e+6ty_e}{1+t}\ge p_e+t(5y_e-4).$$

The numerator of the last difference, divided by t, is $(1-5t)y_e+4+4t-2p_e\ge 0$. After clamping, the lower capacity is $a_e=\kappa+\min(1,u_e+t(5y_e-4))\ge 7\kappa/8$ on active edges, and κ elsewhere. Nonnegativity is needed for uncrossing.

If weighted partition slacks separate the feasible response set from the nonnegative orthant, a minimizing weight distribution can be made a refinement chain. For incomparable partitions, refinement/coarsening decreases the crossing indicator sum, while the sum of part counts increases. Nonnegative capacities and a positive packing target give the correct direction of the slack inequality. Maximizing squared part counts among minimizers rules out incomparable positive weights. Near-tight partitions have slack at most one. Splitting one part and subtracting invariants gives internal capacity at least $Q-1$; removing baselines loses less than one, so their active parts are $\lfloor r/2 \rfloor$-connected once $Q \geq 3r/4$ and r is large. The geometric common response then proves the stronger target $Q - grt$, leaving margin grt for the implemented target $Q - 2grt$.

The number of steps is $O((m+1)(1+\log r))$. The total loss is $O(r^{7/8}(1+\log r))$. At termination $s^4 \geq r$ and active edges have resistance strictly below $r^{-1/4}$. Dropping baselines and then using an integer packing yields $\lfloor Q-1 \rfloor$ trees. For partitions with at least two parts, the subtracted one is at most $|\mathcal{P}|-1$; the one-part case is zero separately. The resulting loss is bounded by $Ar^{15/16}$ for a fixed sufficiently large cutoff.

4.2 An exact separation oracle is not cut enumeration

For distinct vertices set $T_{ijk} = b_{ij}b_{ij}^T + b_{jk}b_{jk}^T - b_{ik}b_{ik}^T$. The source's finite conic representation is

$$\mathcal{N}^* = \{P + \sum \lambda_{ijk} T_{ijk} : P \succeq 0, \lambda_{ijk} \geq 0\}. \tag{4.1}$$

It is closed: $\operatorname{tr} T_{ijk} = 2$ and nonnegative traces bound every coefficient sum and the PSD trace along a convergent sequence. Subsequences then converge in the finite-dimensional coefficient and PSD spaces. Dualizing gives precisely the PSD and triangle inequalities defining $\mathcal{N}$. The trace here is the inherited trace on E, not an unadjusted trace in arbitrary coordinates.

For nonnegative rational z and $q \geq 0$, the partition inequalities are equivalent to

$$z \cdot \ell \geq q \operatorname{MST}(\ell), \qquad 0 \leq \ell \leq 1. \tag{4.2}$$

One direction assigns length one to crossing edges of a partition and zero to the others. Every tree has at least $d-1$ such edges, whether or not the parts are internally connected. Conversely, for components $\mathcal{P}_a$ of edges shorter than a, Kruskal gives $\operatorname{MST}(\ell) = \int_0^1 (|\mathcal{P}_a| - 1)\, da$. A crossing edge at level a has length at least a, so integration gives (4.2). The LP maximizing $qv - z \cdot \ell$, with $v \leq \ell(T)$ for every tree, has an MST separation oracle and rational polynomial-size facet coefficients. Exact rational polyhedral optimization [12] supplies the violated inequality; it need not enumerate partitions. Matrix constraints are separated by rational negative-quadratic-form witnesses from symmetric elimination.

4.3 Interior radius and denominator reset

Write $P = n + m + 2$, use the upper-left block coordinates of a centered matrix, and include scalar variables λ, y, z. The total dimension is at most P^3. A coordinate perturbation of norm η changes the full matrix in operator norm by at most $2P\eta$. A path between extremal coordinates of a centered unit vector gives

$$\lambda_{\min}(L_H|_{\mathsf{E}}) \geq 1/[n(n-1)] \geq P^{-3}, \qquad \|L_H\| \leq 2m.$$

At the auxiliary point $Y_0 = 2L_H$, $\lambda_0 = y_0 = P^{-8}$, $z_0 = \kappa/2$, all nonpartition constraints have a ball of radius P^{-20}. Mixing fraction P^{-8} of it with the stronger response leaves partition margin at least $grt/2$, since $grt \geq 1/(32P)$. Normals of partition rows have norm at most $\sqrt{m}$. Thus a ball of radius P^{-30} satisfies the full system. An outer radius $2P^3$ is valid by trace bounds. The ratio

of radii has polynomial logarithm. Exact membership plus this positive inradius supports a finite rational ellipsoid argument; arbitrary boundary-only SDP feasibility would not.

Lemma 4.1 (Fixed-grid interior rounding)**.** *If a convex set $F \subset \mathbb{R}^D$ contains $x \pm \eta e_j$ for every j, it contains every $x + v$ with $\|v\|_1 \le \eta$. Rounding x coordinatewise to mesh γ remains in F whenever $D\gamma/2 \le \eta$.*

Proof. The indicated $2D$ points have convex hull the ℓ^1 ball of radius η about x. Each rounded coordinate changes by at most $\gamma/2$. □

Taking $L = \lceil \log_2 P \rceil$, $\eta = 2^{-32L}$ and $\gamma = 2^{-36L}$ proves the source's reset bound $D\gamma/2 \le \eta/(2P)$. The translate intersection still contains a ball of radius at least $3P^{-30}/4$. Each returned response coordinate uses at most $39L + O(1)$ bits, independently of the old ramp denominators. The X update adds only the fixed mesh denominator and the denominator of t at each step. Across polynomially many steps its bit length grows additively, not by uncontrolled repeated composition of oracle output lengths. Exact inverses have polynomial bit length by determinant bounds.

5 Normalization without an exact real minimizer

For a union B of q disjoint trees let c_e be reduced incidence columns, $M = q(n-1)$, and

$$N(d) = \sum_e e^{d_e} c_e c_e^T, \qquad f(d) = \log\det N(d) - q^{-1} \sum_e d_e.$$

Cauchy–Binet and total unimodularity of tree minors give $\det N(d) = \sum_T \exp(\sum_{e\in T} d_e)$. Keeping the q specified trees and applying AM–GM gives $f \ge \log q$. The gradient is leverage minus $1/q$, and the Hessian is the covariance of a tree incidence vector under the indicated weighted tree law. Consequently $0 \preceq \nabla^2 f \preceq MI$ and $\|\nabla f\| \le \sqrt{M}$. This law is not the uniform packing mixture.

With gradient error at most $1/(10r^2)$, step $\alpha = 1/[2(M+1)]$, and stopping threshold $\|v\| \le 2/r^2$, a nonstopping step has error at most $\|v\|/20$ and decreases f by at least $\alpha\|v\|^2/2 \ge 2\alpha/r^4$. The initial gap is at most $n\log(2M)$, yielding $O((M+1)r^4 n\log(2M))$ updates and terminal gradient norm at most $3/r^2$. No minimizer at infinity or exact attainment is required.

Proposition 5.1 (Explicit leverage perturbation budget)**.** *Suppose $N = \sum_e w_e c_e c_e^T \succ 0$, $0 < w_e \le W$, $\|c_e\|^2 \le 2$, and $N \succeq \mu I$. If $|\widetilde{w}_e - w_e| \le \delta$ and $2M\delta \le \mu/2$, then*

$$\left|\widetilde{w}_e c_e^T \widetilde{N}^{-1} c_e - w_e c_e^T N^{-1} c_e\right| \le 4\delta/\mu + 8WM\delta/\mu^2.$$

Proof. $\left\|\widetilde{N} - N\right\| \le 2M\delta$ gives $\widetilde{N} \succeq \mu I/2$. The resolvent identity bounds $\left\|\widetilde{N}^{-1} - N^{-1}\right\| \le 4M\delta/\mu^2$. Split the scalar difference into the weight error times $c_e^T\widetilde{N}^{-1}c_e$ and w_e times the inverse difference. The first is at most $4\delta/\mu$, the second at most $8WM\delta/\mu^2$. □

For a gradient-vector error ε_g, it suffices to choose

$$\delta \le \min\left\{\frac{\mu}{4M}, \frac{\varepsilon_g}{M(4/\mu + 8WM/\mu^2)}\right\}.$$

The factor M safely replaces $\sqrt{M}$. On the prescribed iterate box $|d_e| \le B_0$, use $W \le 2^{2\lceil B_0\rceil}$ and $\mu \ge 2^{-2\lceil B_0\rceil}(2M)^{-(n-2)}$. Their logarithms, and hence the needed exponential precision, are polynomial. Rational gradient estimates lie on one fixed dyadic mesh. The iterates then lie on the fixed mesh $[2(M+1)2^{b_0}]^{-1}\mathbb{Z}^M$; repeated updates do not multiply a new denominator at each step.

At termination choose strictly positive rational weights a_e with relative error at most $1/(100r^2)$ from e^{d_e}. Set $N_a = \sum a_e c_e c_e^T$ and choose an exact whitening F with $FN_aF^T = I$. Then $z_e = \sqrt{a_e}Fc_e$ obeys

$$\sum z_e z_e^T = I, \qquad \|z_e\|^2 \le \frac{1 + 1/(100r^2)}{1 - 1/(100r^2)}(q^{-1} + 3/r^2) \le (1 + 4/r)/q.$$

Rational LDL^T followed by approximated positive square roots permits arbitrary polynomial-bit vector accuracy. Exact subset ranks refer to the positive-weight/common-map specification, not to those approximations.

6 Padding and the signing interface

6.1 An exact rational PSD factorization

A rational PSD matrix admits pivoted rational elimination $A = \sum_j d_j u_j u_j^*$, with $d_j > 0$. A zero diagonal row is zero, since $|a_{ij}|^2 \le a_{ii}a_{jj}$. Otherwise a positive pivot produces a PSD Schur complement. Clearing input denominators and applying determinant bounds shows that all pivots and coefficients have polynomial bit length. For $d = a/b > 0$ and $ab = \sum_k c_k 4^k$,

$$d = \sum_k c_k (2^k/b)^2, \qquad c_k \in \{0, 1, 2, 3\}. \tag{6.1}$$

Replacing each pivot by these rational square columns needs only $O(\log(ab))$ columns, not integer factorization or a four-squares oracle. The zero matrix has an empty decomposition, and real data remain real.

6.2 Approximation and removal of the padding

For the stacked exact columns $w_e = (X^{-1/2}b_e, z_e)$, block Cauchy–Schwarz gives $\sum w_e w_e^T \preceq 2I$ and $\|w_e\|^2 \le 5r^{-1/4}$. Approximate $w_e/2$ by rational a_e with error at most $1/(1000|B|r)$. In general, for $\|v\|_e \le R_e$ and $\|a_e - v_e\| \le \delta_e$,

$$\left\|a_e a_e^T - v_e v_e^T\right\| \le 2R_e\delta_e + \delta_e^2, \quad \left\|\sum \sigma_e (a_e a_e^T - v_e v_e^T)\right\| \le \sum_e (2R_e\delta_e + \delta_e^2). \tag{6.2}$$

This follows by expanding the outer product and using the triangle inequality; it holds for every sign vector at once. Here it gives the source's conservative total budget $1/(100r)$ and $\Sigma = \sum a_e a_e^T \preceq 3I/4$. Coordinate error 2^{-b} implies vector error at most $\sqrt{d}\,2^{-b}$, so it is sufficient to take $2^{-b} \le 1/(1000|B|rd)$ in dimension $d \ge 1$, plus guard bits for the already bounded whitening computation.

Factor $I - \Sigma = VV^T$ rationally. If V has columns v_j, add J auxiliary coordinates and, for even $N = 2r$, the N^2 columns $(v_j, \tau_{jk} f_j)/N$, half with each fixed sign τ_{jk}. Their unsigned covariance is $\mathrm{diag}(I - \Sigma, I_J)$, exactly. The maximum squared norm is at most $3r^{-1/4}$ and the number of columns is polynomial because $r \le m$ for explicit graphs.

Let S be any signed augmented covariance with $\|S\| \le T$. Its lower principal block is diagonal, $D_0 = \mathrm{diag}(N^{-2}\sum_k \sigma_{jk})$, and the padding's upper contribution is exactly VD_0V^T. Since $VV^T \preceq I$, $\|D_0\| \le T$ and $\left\|VD_0V^T\right\| \le T$. Hence

$$\left\|\sum_{e \in B} \sigma_e a_e a_e^T\right\| \le 2T. \tag{6.3}$$

No condition on the signed off-diagonal block is asserted or needed. The source's rational tests $TI \pm S \succeq 0$ are exact. Taking dyadic $\sqrt{\Delta} \le d_0 \le 2\sqrt{\Delta}$, $T = 2^{15} d_0$, and applying (6.2) gives

$$\left\| \sum_{e \in A} w_e w_e^T - \tfrac{1}{2} \sum_{e \in B} w_e w_e^T \right\| \le 4T + 2/(100r) \le 2^{20} r^{-1/8}.$$

The generous certificate threshold is not an optimized discrepancy constant or a claim of practical efficiency.

6.3 Quantitative signing: what carries the bit proof

The source appendix normalizes nonzero rank-one Hermitian matrices by $D = \sum |\operatorname{tr} H_i|$ and their trace signs. Then $A_i \succeq 0$, $\sum \operatorname{tr} A_i = 1$ and $1/(Nd^2) \le \nu_0 = \|\sum A_i^2\| \le 1$. Dimension zero and the all-zero list must be returned before this division. Rational PSD bisection supplies $\nu_0 \le \nu \le 1.001\nu_0$, $\varepsilon = \nu/d$, and rational $\sqrt{\nu}/(200N) \le \lambda \le \sqrt{\nu}/(100N)$. Thus $\sigma = \lambda^2/\nu$ is inverse polynomial in N.

The potential is $\Psi = R + \lambda \sum_i (1 - x_i^2)^{1/3}$, where R minimizes $t + \varepsilon \operatorname{tr}(X + Y)$ under

$$X^{-1} + S(x) + \eta_x(Y) \preceq tI, \quad Y^{-1} - S(x) + \eta_x(X) \preceq tI, \qquad \eta_x(Z) = 384 \sum_i \psi_i A_i Z A_i.$$

The source establishes $\|S(x)\| \le R(x)$, a scalar upper bound, strictly positive duals, and a unique minimizer. Its linearized operator $\mathcal{L}$ is self-adjoint for the Frobenius pairing and has a positive inverse with $\mathcal{L}^{-1}(I, I) = (P, Q)/\varepsilon$. Positivity first bounds the inverse in the maximum operator norm; self-adjointness then bounds its Frobenius induced norm by its spectral radius. Omitting this last step would confuse two different matrix norms.

The important uniform estimates are

$$\left\| \mathcal{L}^{-1} \right\| \le \varepsilon^{-1}, \quad \frac{2\varepsilon^2}{100} I \preceq \mathcal{H} \preceq 16(100)^3 I, \quad \langle \mathcal{L}^{-1} E, \mathcal{H} \mathcal{L}^{-1} E \rangle \ge \frac{1}{100d}.$$

They follow respectively from positivity, substituting $P \succeq \varepsilon X^2$ in the inversion Hessian, and $\operatorname{tr}(P + Q) = 1$. Eliminating the scalar multiplier in the full KKT system therefore gives a Jacobian inverse bounded polynomially in d, ε^{-1}. On strips at least $\sigma/2$ from the active coordinate endpoints, three implicit differentiations yield the source's explicit polynomial M bounding second and third derivatives. A finite collection of compact faces alone would not supply this quantitative fact.

The covariance argument gives either an endpoint decrease or $\|\nabla \Psi\|_\infty \ge a$ or $\lambda_{\min}(\nabla^2 \Psi) \le -8a$, with $a = \lambda/(8C_0^2)$ inverse polynomial. The covariance is an existence witness for a direction; the algorithm never has to construct its irrational subspace. Value queries use rational approximations of cube roots and a boxed SDP with a uniform strict interior point. This is weak optimization, not exact optimization of an arbitrary SDP.

Central finite differences use mesh $h \le \min(\sigma/8, 1, a/(64MN))$ and value error $ah^2/(128N)$. The Hessian error is below $a/16$. An exact PSD test on $\widehat{H} + 7aI$ either passes or returns a rational negative direction. Its rounded displacement q has $\|q\| \le s$ and $q^T \nabla^2 \Psi q \le -6as^2$. Both q and $-q$ are tested so their linear terms cancel in the average. With $s \le \min(\sigma/4, a/(4M), 1/4)$, one candidate improves by 2Δ, $\Delta = as^2$. Value error $\Delta/4$ costs at most $\Delta/2$ in choosing the best candidate, leaving $3\Delta/2$ progress. Near-endpoint freezes do not increase the potential and occur at most N times. A fixed dyadic displacement mesh, not continually unrelated rounding, controls coordinate denominators. Finally $(2\sqrt{386} + 0.01)\sqrt{1.001} < 40$.

These calculations identify a finite-bit route conditional on the stated analytic descent inequalities. The checker does not implement the SDP value oracle, the derivative bound or the full walk; no runtime observation is being substituted for that argument.

7 A whitening-free shared-subgraph certificate

The following local result separates *verification of a particular output* from *construction of outputs on every input*. It retains the source's rank argument but eliminates square roots from the final certificate. This is a representation improvement, not a proved speedup.

Theorem 7.1 (Rational cut and packing certificate)**.** *Let H be a connected loopless explicit multigraph, $B \subseteq H$, $A \subseteq B$, and $n \geq 2$. Let c_e be reduced incidence columns, $a_e > 0$ rational on B, and $N = \sum_{e\in B} a_e c_e c_e^T \succ 0$, $N_A = \sum_{e\in A} a_e c_e c_e^T$. Suppose rational data satisfy*

$$X \succeq L_H, \quad 3L_H - X = P + \sum \lambda_{ijk} T_{ijk}, \quad P \succeq 0,\ \lambda_{ijk} \geq 0, \tag{7.1}$$

$$L_A \preceq \theta L_B + eX, \quad N_A \succeq \beta N, \tag{7.2}$$

$$a_e c_e^T N^{-1} c_e \leq \ell \quad (e \in B), \quad \theta, e \geq 0, \quad \beta \geq 0, \quad \ell > 0. \tag{7.3}$$

Then the same *subgraph A satisfies*

$$|\delta_A(S)| \leq (\theta + 3e)|\delta_H(S)| \quad \textit{for all cuts,} \qquad A \textit{ contains } \lfloor \beta/\ell \rfloor \textit{ edge-disjoint spanning trees.}$$

All premises are exactly decidable in polynomial binary time in the encoding length of the graph and displayed certificate.

Proof. For a centered cut indicator x, $xx^T \in \mathcal{N}$, so (7.1) implies $x^T X x \leq 3x^T L_H x$. Equation (7.2), $L_B \preceq L_H$ and $\theta \geq 0$ prove the cut bound for every S without searching for a maximizing cut.

For the packing assertion, define only for the proof $z_e = \sqrt{a_e} N^{-1/2} c_e$. These columns have covariance I, individual squared norms at most ℓ, and the subset A has covariance at least βI. The common invertible map and positive scalings preserve every incidence rank. For a partition into d parts, the span W of columns internal to parts has dimension at most $n - d$. Project onto $W^\perp$ and take traces. The result is

$$\beta(d-1) \leq \sum_{e\in E_A(\mathcal{P})} \|\Pi z_e\|^2 \leq \ell |E_A(\mathcal{P})|.$$

The integer tree-packing criterion now applies; its one-part condition is identically zero. Checking subgraph membership, positive weights, PSD inequalities and the exact inverse uses rational arithmetic. Elimination entries are ratios of minors of denominator-cleared input matrices, hence have polynomial bit length. This also covers singular PSD residuals, with pivoting and zero-row handling. □

The source corresponds to $\theta = 1/2$, $e = C_s r^{-1/8}$, $\beta = 1/2 - e$, and $\ell = (1 + 4/r)/q$; in an actual rational certificate, use rational outward bounds for e, ℓ and an inward bound for β. Exact extraction and frame weights specify the required matrices. The theorem also permits a data-dependent certified β and ℓ in place of coarse universal estimates. This can improve an individual certified packing lower bound, but no worst-case gain follows without new inequalities.

For a still smaller acceptance record, output explicit disjoint trees in A as well as (7.1) and the cut LMI. Then the reviewer can check connectivity and disjointness directly and need not trust the frame construction at all. Conversely a valid frame certificate does not establish that every input admits a response the algorithm can find. Certificate existence and verifier correctness are separate claims.

8 Iteration and compressed inputs

8.1 The loss product

The two blocks of the signed stack prove cut halving and packing retention for exactly one sign set A. A source step has $r/4 \le r' \le r/2$ and

$$a_r = \tfrac{1}{2} + 3C_s r^{-1/8}, \qquad \frac{a_r}{r'/r} \le 1 + C_1 r^{-1/16}.$$

Starting with $r_{\mathrm{init}} = \max(1, \lfloor k/2 \rfloor) \ge k/3$ and stopping below an absolute threshold R_0, the product telescopes:

$$\prod_{i<J} a_{r_i} \le \frac{r_J}{r_{\mathrm{init}}} \exp\left(C_1 \sum_{i<J} r_i^{-1/16} \right), \qquad \sum_{i<J} r_i^{-1/16} \le \frac{R_0^{-1/16}}{1 - 2^{-1/16}}.$$

The last sum starts at the last *processed* parameter, which is at least R_0, not the terminal value below R_0. The empty iteration case uses an empty product and $r_{\mathrm{init}} < R_0$. A permissible explicit constant is any upper bound on $3R_0 \exp(C_1 R_0^{-1/16}/(1 - 2^{-1/16}))$. This expression is not a practical optimized constant. A construction with hardwired universal cutoffs is polynomial only after all geometric and signing constants are justified uniformly.

8.2 Sharper integer connectivity after compression

Proposition 8.1 (Floor-loss transfer)**.** *Let $n \ge 2$, $q = \lfloor n^2/4 \rfloor$, $L \ge q$ an integer, $k \ge 1$, and $c_{uv} \in \mathbb{Z}_{\ge 0}$ with every nontrivial cut of total multiplicity at least k. Define*

$$a_{uv} = \lfloor L \min(c_{uv}, k)/k \rfloor, \qquad K_* = L - q + 1.$$

The explicit auxiliary graph is K_*-edge-connected, has at most $L\binom{n}{2}$ edges, and satisfies $|\delta_A(S)| \le (L/k)|\delta_G(S)|$ for every cut. An explicit C/K_*-thin-tree routine therefore yields an original $(CL/K_*)/k$-thin tree.

Proof. Truncation preserves cut mass at least k: either a crossing pair is truncated and supplies k alone, or nothing in that cut changes. If a cut has d positive crossing pairs, then $1 \le d \le q$ and the sum of its floors is strictly greater than $L - d$. This strict inequality includes integral coordinates because $\lfloor x \rfloor > x - 1$. The cut size is an integer, hence at least $L - d + 1 \ge K_*$. The upper bound follows coordinatewise before summation. A selected auxiliary pair has positive original multiplicity, and a tree uses it at most once, so lifting preserves the cut counts and acyclicity. □

At $L = 2n^2$, $K_* = 2n^2 - \lfloor n^2/4 \rfloor + 1$ and

$$\frac{L}{K_*} \le \frac{8}{7}, \qquad |E(A)| \le 2n^2 \binom{n}{2} < n^4.$$

This replaces the source's factor 2 by at most 8/7 without changing the asymptotic explicit size. It remains conditional on the explicit thin-tree routine. For rational $\epsilon > 0$, choosing $L = q + \lceil q/\epsilon \rceil$ gives $L/K_* \le 1 + \epsilon$ and $O(n^4 + n^4/\epsilon)$ explicit copies. This is polynomial in $1/\epsilon$, not in $\log(1/\epsilon)$. Binary multiplicity arithmetic uses exact products, division with remainder and comparisons; it does not enumerate the original copies.

9 Fractional costs: preservation and an improved constant

The source's fractional input is an explicit list of rational records $(x_e, c_e) \geq 0$ with $x(\delta S) \geq 1$. It is not a real-number oracle. Discard loops. For a nonloop pair p set $X_p = \sum_{e \in E_p} x_e$; if positive, choose a minimum-cost record e_p with $x_{e_p} > 0$ and put $d_p = c_{e_p}$. Then

$$\sum_p d_p X_p \leq \sum_e c_e x_e =: F. \tag{9.1}$$

For $L = 2n^2$, form $a_p = \lfloor L \min(X_p, 1) \rfloor$ labelled copies, each of cost d_p. The same strict floor-loss argument gives K_* connectivity and

$$|\delta_A(S)| \leq L x(\delta S), \qquad c(A) \leq LF, \qquad L/K_* \leq 8/7. \tag{9.2}$$

Theorem 9.1 (Conditional $96B/7$ thin-and-cheap rounding)**.** *Assume a deterministic polynomial-bit explicit routine that finds a C/k-thin tree for every k-edge-connected input. Fix an integer $B \geq \max(1, C)$. For every rational fractional input above on $n \geq 2$ vertices there is a deterministic polynomial-bit construction of one positive-support tree T with*

$$|\delta_T(S)| \leq \frac{96B}{7} x(\delta S) \quad \textit{for every cut,} \qquad c(T) \leq \frac{96B}{7} F.$$

Proof. Write $K = K_*$. If $K < 12B$, any tree of the connected auxiliary graph has cut counts and cost at most those of A. Since $L = (L/K)K < 96B/7$, (9.2) proves both bounds. Otherwise put $k = \lfloor K/2 \rfloor \geq K/3$ and $m = \lfloor K/(6B) \rfloor \geq K/(12B)$. Repeatedly extract a thin tree from the residual graph with parameter k, then delete its labelled copies. After $i \leq m$ deletions, every residual cut is at least

$$(1 - 3Bi/K)|\delta_A(S)| \geq |\delta_A(S)|/2 \geq K/2 \geq k.$$

Indeed each extracted tree uses at most $(B/k)|\delta_{R_i}(S)| \leq (3B/K)|\delta_A(S)|$; induction proves the bound and legality of every call. Each tree is $(3BL/K)$-thin relative to x, at most $24B/7$. The extracted trees are disjoint as labelled auxiliary copies, so a least-cost tree has cost at most $c(A)/m \leq 12BLF/K \leq 96BF/7$. Lift each selected pair to e_p. A single tree uses no pair twice, so cut counts and cost are unchanged and all output records have $x_e > 0$. Different candidate trees may lift to the same original record; only one is returned, so this causes no problem.

There are fewer than n^4 auxiliary edges and $O(n^2)$ calls with an $O(\log n)$-bit connectivity parameter. Pair sums and exact cost comparisons have bit length polynomial in the sum of input bit lengths. No common denominator is expanded into copies. If $F = 0$, all retained positive-support representatives have cost zero by (9.1), and the returned cost is zero; no division by F was used. □

This is a local constant improvement of the reduction, not an independent solution of the fractional rounding problem without its stated routine. The sharper instance-dependent bound $12BL/K$ can also be retained. For $n = 1$, the empty tree gives zero cost and vacuous cut inequalities. Negative costs would invalidate deletion averaging and loop removal; they are outside the theorem.

The source also discusses symmetrized Held–Karp data $z_{uv} = (n-1)(h_{uv} + h_{vu})/n$. Flow balance gives $z(\delta S) \geq 2(n-1)/n \geq 1$. The cheaper orientation of a support pair need not have positive directed h value; the support assertion is about the undirected z pair. The full-manuscript and SODA theorem numbering in the cited historical literature differ. We do not turn this rounding implication into a new ATSP approximation-order claim.

10 Advantages, limits and concrete next steps

The strongest feature of M321's architecture is the explicit connection between one sign set, an all-cut cone inequality and a partition-rank lower bound. The continuation's positive baseline makes uncrossing legitimate; the fixed response grid addresses denominator accumulation; the positive frame weights preserve exact ranks; and the augmented padding has a mathematically justified removal rule. These are stronger claims than an informal invocation of numerical whitening or of MSS.

The main limitations are equally specific. The hierarchy transfer still uses substantial companion accounting. The signing walk relies on nontrivial covariance and differentiated optimization estimates. Very large universal thresholds and high-degree polynomial bounds are consistent with theoretical polynomial time but say little about usable graph sizes. No implementation or benchmark here establishes runtime superiority over earlier algorithms. A rational success certificate is an implication about one output; it does not prove that the construction succeeds universally. No local coordinate change can convert a cut-only bound into an all-vector spectral bound.

The private report remains correct where it distinguishes expectation, simultaneity, spectral sufficiency, exact exponential parity enumeration, and conditional positive domination. Its September 7 literature statement must retain that date rather than be silently promoted to a current assessment after the September public claims. The $k \geq 2$ qualification for the packing average is explicit here. Its internal achievement scale is not the applicability catalogue's L5 scale.

Useful future work is to implement the exact certificate of Theorem 7.1, emit the actual rational frame weights, and compare acceptance with the padded certificate on identical graphs. A full implementation should log maximum numerator/denominator lengths, MST-oracle calls, SDP value precision and accepted descent margins. Benchmarking must include graph families, hardware and reproducible seeds before any performance claim. The hierarchy transfer should be audited lemma-by-lemma against its companion's actual scalar invariants. Formal work should target the rational certificate and integer floor lemma separately from existence, then connect them through an explicit algorithmic specification. These are testable tasks, not claims of already completed validation.

11 Reading and evidence ledger

11.1 Source coverage

The common public path is the pinned repository's `preprints/A-polynomial-time-construction-of-strong-thin-trees-September-23-2026/build/source`. The ledger uses file-relative locators and actual source labels rather than inferred printed numbering. Every listed source was read in full.

File	Mathematical reading scope
`main.tex`, `macros.tex`	Complete entry point, section inclusion order, notation and theorem environments.
`sections/introduction.tex`, `sections/main-theorem.tex`	Input models, all-cut quantifiers, literature claims, n=1 return.
`sections/tree_geometry.tex`	Cone, conic dual, doubled profiles, every geometric-transfer case, parameter exponents and simultaneous response. Imported companion proofs remain dependencies.

File	Mathematical reading scope
sections/extraction.tex	Entire ramp, chain uncrossing, exact MST separation, conic representation, inner/outer radii, translated feasibility and grid reset.
sections/normalization.tex	Log determinant, inexact descent, rational positive weights, spectral bounds and whitening precision.
sections/signing.tex	Rational PSD decomposition, stacked approximation, all padding columns, padding removal and exact PSD acceptance.
sections/signing_interface.tex	Entire potential/duality, covariance and drift estimates, uniform Jacobian and derivatives, value oracle, rational negative direction, grid and termination.
sections/iteration.tex	Same-subgraph cut/packing derivation, floors, cutoff and product, binary truncation, edge lift.
sections/fractional_rounding.tex	All proof branches, zero costs, pair representatives, residual induction, bit costs and historical comparison.
figures/metric_geometry.tex, refs/sources.bib	Diagram meaning and all bibliography entries, including exact public signing version.

The directly cited private source is res-math/TR-item-224-astra.tex, under the supplied confidential res-rt root. Full reading includes its revision sections, complete coarea proof, parity transform, positive domination, stability statement and bibliography. The corpus maps were used as locators: core E5/E6/E14, foundations F4/F5 and applications K were inspected for relevance. Actual supporting passages read were TR248 lines 220–266 and 351–403, and TR250 lines 176–237 in the preferred TechReps/FinalTechReps/TRs editions. Those are selected theorem readings, not whole-report audits of TR248 or TR250; no additional result is imported from their unread sections or from the graph/arithmetic map leads.

11.2 Hashes, public refresh and formalization

The accompanying evidence/source-verification.json compares all thirteen assigned source files with the acquisition manifest _sources/source-hashes.json, records SHA-256 values and line counts, and records hashes of the private report, supporting sources, maps and snapshot metadata. The local public directory is an acquired snapshot, not a Git worktree; a failed gitrev-parseHEAD is not a pin verification. Matching the acquisition manifest establishes byte consistency with that local pinned inventory, not an independent proof of the remote repository's provenance.

Public URLs opened on October 7 include Ezeunala–Jiang v1 (the theorem, algorithm and Section 4.1 passages), Anari–Oveis Gharan v4's abstract, the laminar-paper abstract, the near-minimum paper and its theorem, Terao's official Dagstuhl record, and Bansal's official journal record. The bibliography distinguishes public sources from confidential local ones. No private text, file or private report title was uploaded or used as a public search query.

The pinned local formalization-2026-10-07.yaml was inspected. It lists the existence companion and OAI.StrongThinTree.strongThinTree in OAI/Combinatorics/ThinTrees/Main.lean, with ComparatorChallenges/StrongThinTree.json. It does not supply a separate M321 polynomial-bit construction entry. Catalogue inspection is not a reading of that Lean proof, not a replay, and not evidence that the exact signing/extraction implementation has been formally verified.

11.3 Exact checks and their limits

The standalone `evidence/check_m321.py` uses integer and rational arithmetic and records each family count in `evidence/test-results.json`. The completed run passed 102,413 assertions, including thirteen public source-hash comparisons and a source-count assertion. It accepted 344 small packing-certificate cases and examined 3,247 fractional graphs. It checks base-four pivot splitting; rational PSD decomposition including zero pivots; unsigned padding completion and signed removal identities; the ramp lower bound on a finite rational grid; refinement/coarsening uncrossing for all partitions through five vertices; the shared packing certificate on enumerated small weighted graphs; the explicit leverage perturbation bound; the fixed-grid inequalities; and the improved fractional reduction on a declared finite family. No floating-point tolerance is used for these checks. The test artifact is the authoritative count record, and any failed assertion aborts the run.

A separate independent binary-reduction audit and its exact script were also read: `_audit/M321-parent-binary-reduction.txt` and `_audit/check_M321_parent_binary.py`. Its inspected JSON reports 3,891 graphs, 328,416 cut/scale cases and 1,410,581 passing assertions, including 100-digit multiplicities and rational accuracy examples. That run was not re-executed here and its counts are not included in the 102,413 assertions above. Proposition 8.1 integrates the proved strict floor-loss refinement without a historical novelty claim. Both finite test collections corroborate local algebra; neither supplies the universal construction premise.

These checks are not a full thin-tree construction, an exhaustive graph search at unbounded size, an implementation of the ellipsoid method, an SDP solver test, a proof of the hierarchy transfer, or a verification of the signing walk's derivatives. The report's local theorems have proofs independent of the finite test counts. The source's prior test counts in private reports were not rerun and are not credited here. Final source/PDF and evidence hashes are frozen in `evidence/frozen-hashes.json`; compilation and all-page visual review are recorded separately without modifying the originals.

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